University of kasdi Merbah Ouargla
Faculty of New Technologies of Information and Communication
Department of computer science and Information Technology

Module: Electronics, system components

Corrected series of TD n° 1

Exercise 1:

Give the correct answer (A, B, C or D):

1. c
2. b
3. c
4. c
5. A
6. b
7. a, b

Exercise 2:

1. Low current refers to the current used to carry information. Example: telephony, internet, intercoms, etc.

High current refers to the current used to transport electrical energy. Example: lighting (architectural, technical, security, etc.), electrical outlets, electrical appliances, etc.

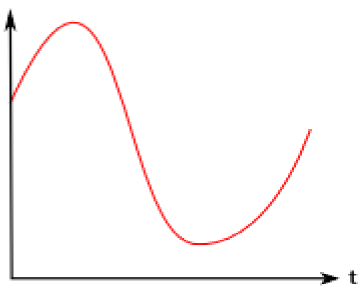
2. The signal carrying information must pass through a transmission channel between a transmitter and a receiver. The signal is rarely suitable for direct transmission through the chosen communication channel. Modulation can be defined as the process by which the signal is transformed from its original form into a form suitable for the transmission channel, for example by varying the amplitude and frequency parameters. (Modulation is generally intended to change the physical form of a signal and, in particular, to place it in the desired frequency band for its transmission or processing).

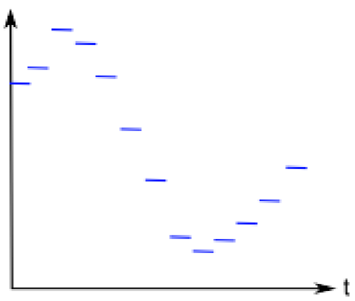
3. Analog signals: these are signals that vary continuously over time. These signals most often reflect variations in physical quantities, for example, temperature, speed, sound, effort,…

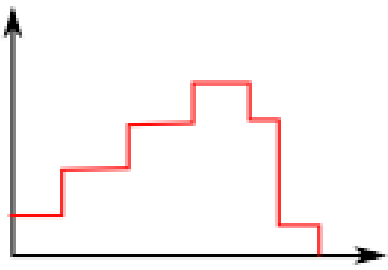
Digital signals: these are discontinuous signals over time. It is a succession of 0s and 1s, called bits. It is said to be binary

4. Analogue: A, D, F

Numeric: B, C, E, G







Change in temperature over the course of a day	Hourly temperature Display	Film recorded by a VCR on VHS tape	Film streamed online

Exercise 3:

1. b) Hertz

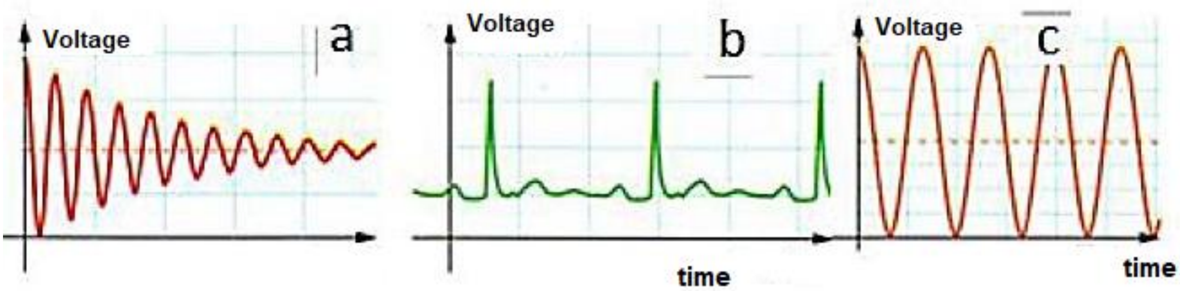
2. a) second

3. d) the inverse of the frequency

4. b) The period of B is longer than that of A

Exercise 4:

1. Period T: it represents the time it takes for the signal to “return to its starting point”. This is the duration of one signal cycle.
Frequency F: is the number of cycles per second, $F = 1 / T$.
2. A phenomenon reproduces twice per second, that is to say F = 2 Hz.
3. The repetitive phenomenon lasts 3 s, this is its period: T = 3 s.
4. Period is the reciprocal of frequency so: $T = 1 / F = 1 / 1.33 = 0{,}75\ Hz$
5. A periodic phenomenon repeats itself identical to itself at a regular time interval T (period).
(b) and (c) periodic because the voltage variations repeat themselves identically. Signal (a) non-periodic because the amplitude decreases



Exercise 5:

1. 2,5 x TA = 10 ms
TA = 4,0 ms = $4{,}0 \times 10^{-3}$ s
2 x TB = 500 us
TB = 250 us = 250×10^{-6} s = 2.5×10^{-4} s
2.

$$F_A = \frac{1}{T_A} = \frac{1}{4x10^{-3}} = 250\ Hz = 0.25\ Khz$$

$$F_B = \frac{1}{T_B} = \frac{1}{2{,}5x10^{-4}} = 0.4x10^4\ Hz = 4x10^3\ \text{HZ} = 4\ \text{Khz}$$

3. He cannot hear sound at a frequency of 250 Hz well; therefore he cannot hear low frequencies well.

Exercise 6:

1. Nombre de caractères transférés par minute : $\frac{56000 \times 60}{8} = 420000$

2. 1 octet = 8 bits
   1 GO = 10^9 octets

1,5 GO = 1,5 x 10^9 octets = 1.5 x 10^9 x 8 = 12 x 10^9 bits
25x 60 = 1500s

$$\text{Speed} = \frac{\text{Quantity}}{\text{Time}} = \frac{12 \times 10^9}{1500} = 8 \times 10^6 \text{ bits/s} = 8 \text{ Mbits/s.}$$

Exercise 7:

1) Active electronic components define the components that increase the power of an electrical signal (current, voltage or both). However, passive electronic components define electronic components that do not increase the power of an electrical signal
2)

Components	Answer
1. Vacuum Tubes,	A
2. Transistors,	A
3. Diodes,	A
4. Integrated Circuit (IC),	A
5. LED light-emitting diode (Light-Emitting Diode)	A
6. Resistance,	B
7. Capacitor,	B
8. Transformer,	B

Good luck
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